BAMBU STAND: 3D PRINTER WORKBENCH BLUEPRINTS

Solid Walnut Top (24"x48"x1") • L-Shaped 2x4 SPF Legs • 3-Tier Spool Drawers • CR-6 SE Bay • Anti-Racking Shear Wall

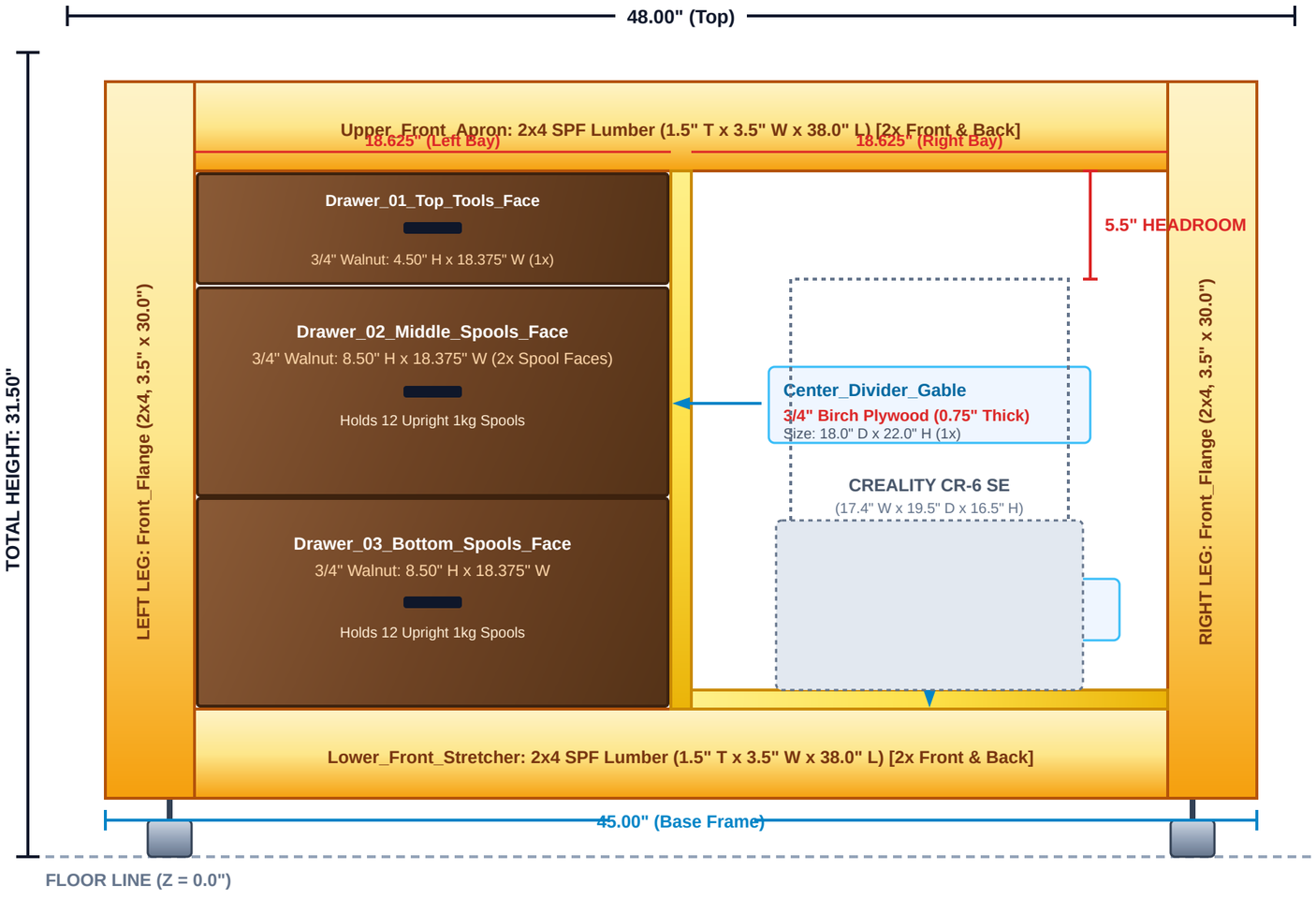
SHEET 1 OF 4: BENCH ELEVATIONS & COMPONENT LABELS

SCALE: NTS (DIMENSIONED) • UNITS: INCHES (")

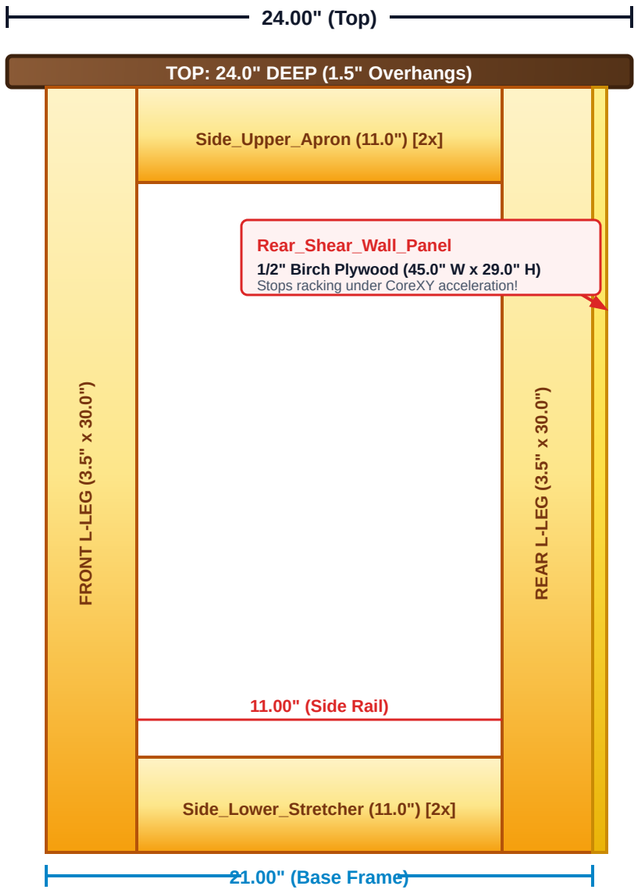
» Bench Elevations & Component Labels: Front elevation, side elevation, bay openings, clearances, and labeled parts

1. FRONT ELEVATION (LOOKING SOUTH INTO WORKBENCH)

OVERALL: 48.0" W x 31.5



2. SIDE ELEVATION (LOOKING WEST)



MASTER LUMBER CUT SCHEDULE & REPEAT COUNTS (ALL BOARDS & PANELS)

COMPONENT LABEL	QTY	MATERIAL SPECIFICATION	FINISHED SIZE (T x W x L)	REPEAT / JOINERY NOTES	SOURCE / SKU
Walnut_Tabletop	1x	1.0" Solid American Walnut	**1.00" x 24.00" x 48.00"**	Existing desktop; 1.5" perimeter overhang all 4 sides	Existing Solid Slab
Front_Rear_Long_Rails	4x	2x4 SPF Premium Stud Lumber	**1.50" x 3.50" x 38.00"**	**REPEAT 4x:** Upper Front, Upper Rear, Lower Front, Lower Rear rails	Home Depot SKU 1000112108
Side_Short_Rails	4x	2x4 SPF Premium Stud Lumber	**1.50" x 3.50" x 11.00"**	**REPEAT 4x:** Upper Left, Upper Right, Lower Left, Lower Right side rails	Home Depot SKU 1000112108
Center_Bay_Cross_Ties	2x	2x4 SPF Premium Stud Lumber	**1.50" x 3.50" x 18.00"**	**REPEAT 2x:** Upper Mid Rail & Lower Mid Rail (anchors divider gable)	Home Depot SKU 1000112108
Corner_L_Leg_Flanges	8x	2x4 SPF Premium Stud Lumber	**1.50" x 3.50" x 30.00"**	**REPEAT 8x:** 4 Front/Rear Flanges + 4 Side Return Flanges (joined at 90°)	Home Depot SKU 1000112108
Corner_Foot_Mount_Blocks	4x	2x4 SPF Lumber Scrap	**1.50" x 2.00" x 2.00"**	**REPEAT 4x:** Glued in inside corner of each L-leg to mount 3/8" insert	From 2x4 offcuts
Center_Divider_Gable	1x	**3/4" PureBond Birch Plywood**	**0.75" x 18.00" x 22.00"**	Vertical bay partition; anchors right slide rails and CR-6 tech shelf	Home Depot SKU 1001513690
CR6_Tech_Shelf	1x	**3/4" PureBond Birch Plywood**	**0.75" x 18.625" x 18.00"**	Heavy lower printer shelf; supports Creality CR-6 SE with 5.5" clearance	Home Depot SKU 1001513690
Rear_Shear_Wall_Panel	1x	**1/2" PureBond Birch Plywood**	**0.50" x 45.00" x 29.00"**	**SCREWED TO REAR FRAME:** Completely stops racking under CoreXY acceleration	Home Depot SKU 1001513691
Spool_Drawer_Faces	2x	**3/4" PureBond Walnut Plywood**	**0.75" x 18.375" x 8.50"**	**REPEAT 2x:** Middle & Bottom Drawer faces; flush mounted against box front	PureBond Walnut 2x4' Panel

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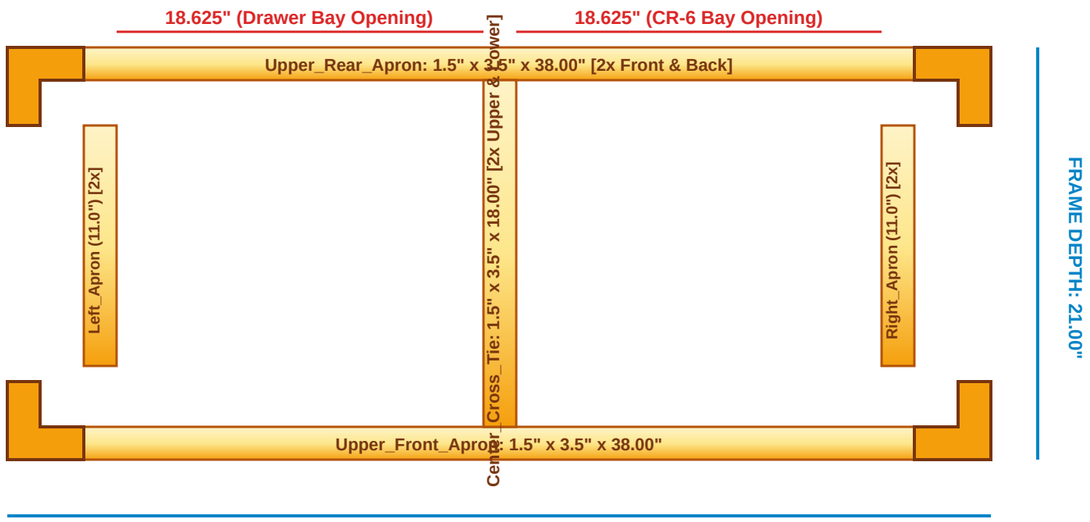
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SHEET 2 OF 4: 2X4 SOLID TIMBER BASE & JOINERY SCHED

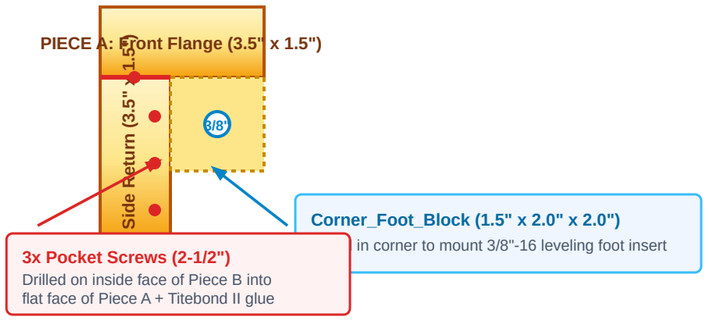
SCALE: NTS (DIMENSIONED) • UNITS: INCHES (")

» 2x4 Solid Timber Base & Joinery Schedule: Upper aprons, lower stretchers, cross-ties, and L-shaped leg assembly breakdown

1. UPPER APRON FRAME PLAN (TOP-DOWN VIEW AT Z = 27.0")



2. DETAIL A: "L-SHAPED" 2x4 CORNER LEG JOINT (ZERO PLANING NEEDED)



WHY THE "L-SHAPED" 2x4 CORNER LEG DESIGN IS SUPERIOR:

- **Zero Planer or Jointer Needed:** Factory-flat edge of Piece B joins factory-flat face of Piece A.
- **No Green Pressure-Treated 4x4s:** Home Depot Canada only stocks green wet 4x4s outdoors. 2x4 studs are dry!
- **Huge Torsional Rigidity:** An L-bracket column has a 3.5" x 3.5" perimeter envelope that resists twist!
- **Internal Corner Foot Pad:** The glued 2x2 corner block gives a solid 1.5" thick base for threaded leveling glides.

3. COMPLETE 2x4 SPF LUMBER CUTTING SCHEDULE & BOARD BREAKDOWN

HOW TO CUT 7 STANDARD 8-FOOT (96") 2x4 STUDS WITH ZERO WOOD WASTE:

STUD #1 (96"):	Long Rail: Upper_Front_Apron (38.00")	Long Rail: Upper_Rear_Apron (38.00")	20" Offcut (Foot blocks)				
STUD #2 (96"):	Long Rail: Lower_Front_Stretcher (38.00")	Long Rail: Lower_Rear_Stretcher (38.00")	20" Offcut				
STUD #3 (96"):	Center Cross-Tie (18.0")	Center Cross-Tie (18.0")	Side Rail (11.0")	Side Rail (11.0")	Side Rail (11.0")	Side Rail (11.0")	16" Offcut
STUDS #4-7 (4x):	L-Leg Piece A: Front/Rear Flange (30.00")	L-Leg Piece B: Side Return Flange (30.00")	3rd Leg Piece / Spare (30.00")				

PART NAME & LABEL	REPEAT QTY	CUT SIZE (T x W x L)	FUNCTION & PLACEMENT IN FRAME
Front_Rear_Long_Rails	4x	1.50" x 3.50" x 38.00"	Upper long rails (2x) & lower long stretchers (2x) spanning front and rear
Side_Short_Rails	4x	1.50" x 3.50" x 11.00"	Upper side rails (2x) & lower side stretchers (2x) connecting front and rear legs
Center_Bay_Cross_Ties	2x	1.50" x 3.50" x 18.00"	Upper & lower cross-ties between front and rear 38" rails; anchors center partition
Leg_Front_Rear_Flanges	4x	1.50" x 3.50" x 30.00"	Front face of the 4 L-shaped corner columns; 30.0" tall for 31.5" finished bench height
Leg_Side_Return_Flanges	4x	1.50" x 3.50" x 30.00"	Side returns of the 4 L-shaped corner columns; pocket screwed to Front Flanges at 90°
Corner_Foot_Mount_Blocks	4x	1.50" x 2.00" x 2.00"	Scrap 2x4 blocks glued inside bottom corner of each L-leg; bored for 3/8" leveling foot insert

TOTAL 2x4 SPF LUMBER NEEDED: Exactly 7 Standard 8-foot (96") Studs (~$45.50 CAD at Home Depot Canada SKU 1000112108)

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SHEET 3 OF 4: PLYWOOD PANELS & THICKNESS GUIDE

SCALE: NTS (DIMENSIONED) • UNITS: INCHES (")

» Plywood Panels & Thickness Guide: Dimensioned cutting layouts, exact thickness specifications, and Home Depot Canada SKUs

PLYWOOD THICKNESS SPECIFICATION LEGEND (COLUMBIA FOREST PRODUCTS PUREBOND):

- **3/4" (0.75") BIRCH PLYWOOD** SKU 1001513690 ($41)
- **1/2" (0.50") BIRCH PLYWOOD** SKU 1001513691 ($36)
- **3/4" (0.75") WALNUT PLYWOOD** Walnut 2x4' ($54)
- **1/4" (0.25") BIRCH PLYWOOD** SKU 1001513692 ($28)

1. 3/4" (0.75") BIRCH PLYWOOD PANELS (SKU 1001513690 - 1 Panel 2'x4')



2. 3/4" (0.75") WALNUT DRAWER FACES (PureBond Walnut - 1 Panel 2'x4')



3. 1/2" (0.50") BIRCH PLYWOOD (SKU 1001513691 - 3 Panels 2'x4') & 1/4" BOTTOMS



DRAWER BOX CUT LIST (1/2" BIRCH WALLS + 1/4" BIRCH BOTTOMS):

DRAWER BOX COMPONENT	QTY	THICKNESS	CUT SIZE (W x H)	DRAWER ASSIGNMENT & NOTES
Spool_Box_Sides	4x	1/2" (0.50")	**18.00" x 7.75"**	**REPEAT 4x:** Left/Right sides for Drawers 2 & 3
Spool_Box_Front_Back	4x	1/2" (0.50")	**17.375" x 7.75"**	**REPEAT 4x:** Front/Back sub-panels for Drawers 2 & 3
Tool_Box_Sides	2x	1/2" (0.50")	**18.00" x 3.50"**	**REPEAT 2x:** Left/Right sides for Top Tool Drawer
Tool_Box_Front_Back	2x	1/2" (0.50")	**17.375" x 3.50"**	**REPEAT 2x:** Front/Back sub-panels for Top Tool Drawer
Drawer_Bottom_Panels	3x	1/4" (0.25")	**17.375" x 17.50"**	**REPEAT 3x:** 1/4" Birch bottoms for all 3 drawer boxes

JOINERY METHOD FOR 1/2" BIRCH DRAWER BOXES:

- Set Kreg Jig to **1/2" material thickness**. Drill pocket holes on outside of front and back sub-panels.
- Fasten using **1" pan-head pocket screws** + Titebond II wood glue. (Completely hidden once walnut face is mounted!).
- Slide rails mount to sides with 1/4" standoff block; 3/4" Walnut faces screw from inside box with 1" screws.

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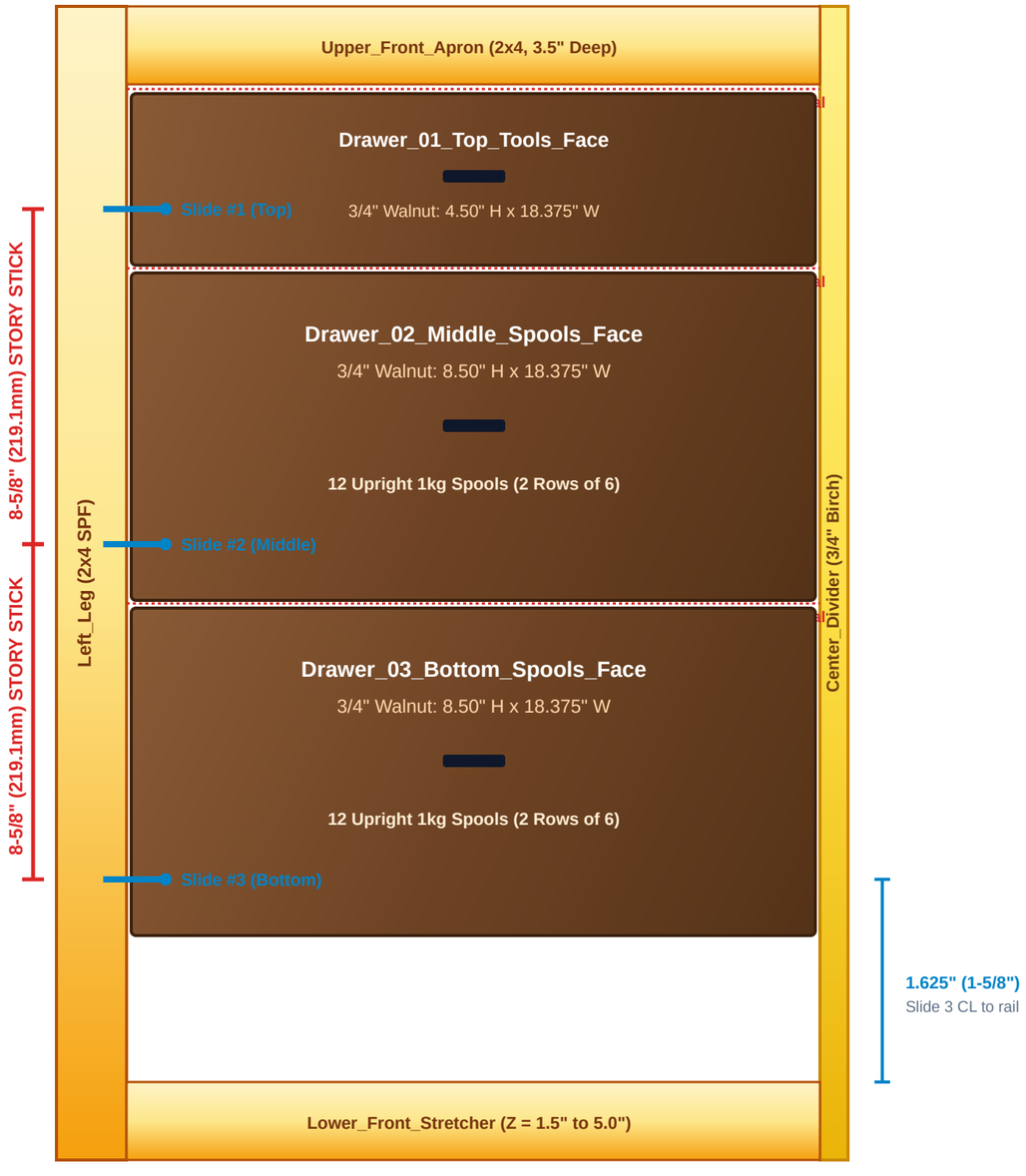
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SHEET 4 OF 4: 3-TIER DRAWER BANK & SLIDE ELEVATIONS

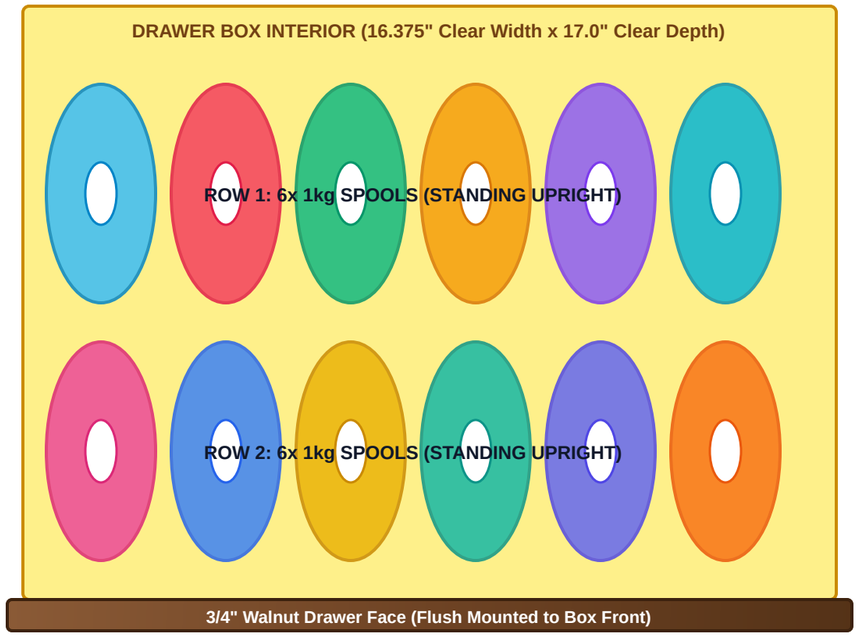
SCALE: NTS (DIMENSIONED) • UNITS: INCHES (")

» 3-Tier Drawer Bank & Slide Elevations: Slide mounting heights, 8-5/8 inch story stick spacing, 1/8 inch reveals, and 12-spool capacity

1. DRAWER STACK ELEVATION & SLIDE TRACK ELEVATIONS



2. SPOOL DRAWER BOX INTERIOR (12 SPOOLS STANDING UPRIGHT)



TOTAL SPOOL STORAGE: 2 DRAWERS x 12 SPOOLS = 24 FULL 1KG FILAMENT SPOOLS!

3. JIG REFERENCE & HARDWARE MOUNTING SUMMARY

HOW THE 3D PRINTABLE JIGS ACCELERATE THIS DRAWER BUILD:

JIG 1: Setback Cradle (drawer_installation_jigs.stl)

- Hooks over the front 2x4 leg/post to automatically set the **2.5mm (3/32") slide setback**.
- Shelf supports the 18" metal slide hands-free; drive screw through the **center access window**.

JIG 2: Mounting Blocks (Pair under drawer box)

- Sits flat on bench under drawer box; supports slide on the **1/4" (6.35mm) standoff step**.
- Lifts slide so bottom screws never scrape, then pinch the **top diamond notch** to pull blocks out.

JIG 3: 1/8" Reveal Shims (Set of 4 around drawer face)

- Slipped into the perimeter gaps around the 3/4" Walnut drawer faces while clamping from inside.
- Guarantees a perfectly uniform **1/8" reveal margin** around all drawers.

SLIDE TIER STORY STICK: 8-5/8" (219mm) Carcass Spacing (slide_tier_story_stick.stl)

- Hooks under lower slide track and holds upper slide track at exact **8-5/8" (219.1mm) elevation**.
- Guarantees Slide 2 and Slide 1 are dead level and identical on both sides of the cabinet.

DRAWER PULL TEMPLATE: 5" (128mm) Bar Pulls (drawer_pull_jig.stl)

- Center notch lines up with 9-3/16" pencil mark; guides 3/16" holes at 128mm spacing.